12TH CHEMISTRY · CHAPTER 22 · FULL CHAPTER MCQS

Hydroxy Compounds

106 topic-wise MCQs from the complete chapter with answers and reasons

INTRODUCTION AND 22.1 ALCOHOLS

1. The functional group of all alcohols is:

(a) –CHO (b) –OH
(c) –COOH (d) –X

Answer: (b) –OH — The hydroxyl group.

2. In phenols the hydroxyl group is attached to:

(a) An aryl carbon (b) An alkyl carbon
(c) A carbonyl carbon (d) A nitrogen atom

Answer: (a) An aryl carbon — Phenols have –OH bonded to an arene.

3. Ethanol is also called:

(a) Wood spirit (b) Glycol
(c) Carbolic acid (d) Grain alcohol

Answer: (d) Grain alcohol — Ethyl alcohol or grain alcohol.

4. Ethanol produced from bio-based materials is used as a:

(a) Disinfectant (b) Dye
(c) Biofuel (d) Fertilizer

Answer: (c) Biofuel — It is used as a biofuel.

5. Phenol is an ingredient of:

(a) Disinfectants (b) Fertilizers
(c) Biofuels (d) Refrigerants

Answer: (a) Disinfectants — E.g. soaps and hospital disinfectant liquids.

6. Scientists started cleaning surgical instruments with phenol solution in:

(a) 1767 (b) 1967
(c) 1905 (d) 1867

Answer: (d) 1867 — After Louis Pasteur’s research on bacteria.

7. Phenol was replaced as an antiseptic because it:

(a) Was too expensive (b) Burns healthy tissues
(c) Does not kill bacteria (d) Is insoluble

Answer: (b) Burns healthy tissues — It was too potent for this use.

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8. The existence of bacteria was revealed by the research of:

(a) Kekulé (b) Wöhler
(c) Louis Pasteur (d) Faraday

Answer: (c) Louis Pasteur — Stated in the chapter introduction.

9. The general formula of alcohols is:

(a) ROH (b) RX
(c) RCHO (d) ROR

Answer: (a) ROH — R is an alkyl or aryl group.

10. The first member of the monohydric alcohols is:

(a) Ethanol (b) Methanol
(c) Propanol (d) Glycol

Answer: (b) Methanol — CH_3OH.

11. An alcohol with only one –OH group is called:

(a) Polyhydric (b) Tertiary
(c) Dihydric (d) Monohydric

Answer: (d) Monohydric — E.g. ethanol.

12. Glycol is:

(a) 1,2,3-Propanetriol (b) Propan-2-ol
(c) 1,2-Ethanediol (d) Ethanol

Answer: (c) 1,2-Ethanediol — A polyhydric alcohol with two –OH groups.

13. Glycerol contains how many hydroxyl groups?

(a) One (b) Four
(c) Two (d) Three

Answer: (d) Three — It is 1,2,3-propanetriol.

14. Which is a polyhydric alcohol?

(a) Propan-1-ol (b) Glycerol
(c) Ethanol (d) Methanol

Answer: (b) Glycerol — It has more than one –OH group.

15. Propan-2-ol is a:

(a) Secondary alcohol (b) Primary alcohol
(c) Tertiary alcohol (d) Polyhydric alcohol

Answer: (a) Secondary alcohol — Its –OH carbon is attached to two carbons.

16. The general formula of a tertiary alcohol is:

(a) RCH_2OH (b) R_2CHOH
(c) R_3COH (d) ROR

Answer: (c) R_3COH — Three alkyl groups on the –OH carbon.

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17. Which is a primary alcohol?

(a) Propan-2-ol
(b) Butan-2-ol
(c) 2-Methylpropan-2-ol
(d) Ethanol

Answer: (d) Ethanol — CH_3CH_2OH has the $-CH_2OH$ group.

18. 2-Methylpropan-2-ol is a:

(a) Primary alcohol
(b) Secondary alcohol
(c) Tertiary alcohol
(d) Diol

Answer: (c) Tertiary alcohol — Its –OH carbon is attached to three carbons.

19. Cyclopentanol is a:

(a) Secondary alcohol
(b) Primary alcohol
(c) Tertiary alcohol
(d) Phenol

Answer: (a) Secondary alcohol — The –OH carbon is bonded to two ring carbons.

20. Which family also contains hydroxyl groups?

(a) Alkanes
(b) Carbohydrates
(c) Alkenes
(d) Halogenoalkanes

Answer: (b) Carbohydrates — Cholesterol and carbohydrates belong to this family.

22.2 ACIDITY OF ALCOHOLS AND PHENOLS

21. The correct order of acidity is:

(a) Ethanol > water > phenol
(b) Phenol > ethanol > water
(c) Water > phenol > ethanol
(d) Phenol > water > ethanol

Answer: (d) Phenol > water > ethanol — Given in the textbook.

22. Alcohols are weaker acids than water because the alkyl group:

(a) Withdraws electrons
(b) Is polar
(c) Donates electrons
(d) Is acidic

Answer: (c) Donates electrons — The positive inductive effect destabilises the alkoxide ion.

23. The ion formed when an alcohol loses a proton is the:

(a) Phenoxide ion
(b) Alkoxide ion
(c) Hydroxide ion
(d) Carbocation

Answer: (b) Alkoxide ion — RO^-.

24. In the dissociation of an alcohol in water the equilibrium lies:

(a) To the left (b) To the right
(c) In the middle (d) Nowhere

Answer: (a) To the left — Very few molecules dissociate.

25. The electron-donating effect of alkyl groups is called the:

(a) Mesomeric effect (b) Positive inductive effect
(c) Negative inductive effect (d) Steric effect

Answer: (b) Positive inductive effect — +I effect.

26. The conjugate base of phenol is the:

(a) Phenoxide ion (b) Alkoxide ion
(c) Phenyl cation (d) Hydroxide ion

Answer: (a) Phenoxide ion — $C_6H_5O^-$.

27. The phenoxide ion is stable because its negative charge is:

(a) Localised on oxygen (b) Removed by water
(c) Delocalised over the ring (d) On hydrogen

Answer: (c) Delocalised over the ring — Less charge density on the oxygen atom.

28. In phenol the lone pair of oxygen overlaps with the:

(a) σ bonds of the ring (b) C–H bonds
(c) O–H bond (d) Delocalised π system of the ring

Answer: (d) Delocalised π system of the ring — This delocalisation makes phenol acidic.

29. A more stable conjugate base means the acid is:

(a) Weaker (b) Neutral
(c) Stronger (d) Unchanged

Answer: (c) Stronger — The proton is lost more readily.

30. Compared with water, the solution of an alcohol has:

(a) A higher $[H^+]$ (b) No H^+ at all
(c) The same $[H^+]$ (d) A lower $[H^+]$

Answer: (d) A lower $[H^+]$ — So alcohols are weaker acids.

31. More electron density on the oxygen of an alkoxide ion makes it:

(a) More stable (b) Less stable
(c) Neutral (d) Coloured

Answer: (b) Less stable — It accepts H^+ more readily.

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32. The weakest acid is:

(a) Ethanol (b) Phenol
(c) Water (d) Carbolic acid

Answer: (a) Ethanol — Its ethoxide ion is the least stable.

33. Which reacts with aqueous NaOH?

(a) Ethanol (b) Phenol
(c) Methanol (d) Propan-2-ol

Answer: (b) Phenol — Only phenol is acidic enough.

34. Sodium gives hydrogen more rapidly with:

(a) Ethanol (b) Butan-1-ol
(c) Propan-1-ol (d) Phenol

Answer: (d) Phenol — Phenol is the stronger acid.

22.3 SYNTHESIS OF ALCOHOLS

35. The catalyst for the reaction of steam with an alkene is:

(a) Conc. H_2SO_4 only
(b) $ZnCl_2$
(c) Al_2O_3
(d) Conc. H_3PO_4

Answer: (d) Conc. H_3PO_4 — Concentrated phosphoric acid.

36. The hydration of ethene is an example of:

(a) Electrophilic addition (b) Nucleophilic substitution
(c) Elimination (d) Free-radical substitution

Answer: (a) Electrophilic addition — Stated in the textbook.

37. Ethene with cold dilute potassium manganate(VII) gives:

(a) Ethanol (b) Ethane-1,2-diol
(c) Ethanal (d) Ethanoic acid

Answer: (b) Ethane-1,2-diol — A diol is formed.

38. Cold dilute $KMnO_4$ is a:

(a) Strong reducing agent (b) Dehydrating agent
(c) Mild oxidising agent (d) Catalyst

Answer: (c) Mild oxidising agent — It oxidises alkenes to diols.

39. Bromoethane heated with aqueous NaOH gives:

(a) Ethene (b) Ethane
(c) Ethanal (d) Ethanol

Answer: (d) Ethanol — Nucleophilic substitution of Br by OH.

40. The conversion of a halogenoalkane into an alcohol is:

(a) Electrophilic addition (b) Nucleophilic substitution
(c) Elimination (d) Oxidation

Answer: (b) Nucleophilic substitution — OH^- replaces the halogen.

41. Reduction of an aldehyde gives a:

(a) Primary alcohol (b) Secondary alcohol
(c) Tertiary alcohol (d) Ketone

Answer: (a) Primary alcohol — E.g. ethanal → ethanol.

42. Reduction of a ketone gives a:

(a) Primary alcohol (b) Tertiary alcohol
(c) Secondary alcohol (d) Carboxylic acid

Answer: (c) Secondary alcohol — E.g. propanone → propan-2-ol.

43. Reduction of propanone gives:

(a) Propan-2-ol (b) Propan-1-ol
(c) Propanal (d) Propanoic acid

Answer: (a) Propan-2-ol — A secondary alcohol.

44. Carboxylic acids are reduced to primary alcohols by:

(a) $NaBH_4$ only
(b) $LiAlH_4$
(c) $K_2Cr_2O_7$
(d) $KMnO_4$

Answer: (b) $LiAlH_4$ — Given in the textbook.

45. Reduction of ethanoic acid gives:

(a) Ethanal only (b) Ethene
(c) Propan-2-ol (d) Ethanol

Answer: (d) Ethanol — A primary alcohol.

46. Which are reducing agents for aldehydes and ketones?

(a) $K_2Cr_2O_7$ and $KMnO_4$
(b) H_3PO_4 and H_2SO_4
(c) $NaBH_4$ and $LiAlH_4$
(d) $NaNO_2$ and HCl

Answer: (c) $NaBH_4$ and $LiAlH_4$ — Named in the textbook.

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47. Hydrolysis of an ester gives:

(a) An aldehyde and a ketone
(b) A carboxylic acid and an alcohol
(c) Two alcohols
(d) An alkene and water

Answer: (b) A carboxylic acid and an alcohol — On heating with dilute acid or alkali.

48. Hydrolysis of ethyl ethanoate gives ethanoic acid and:

(a) Methanol (b) Ethanal
(c) Propanol (d) Ethanol

Answer: (d) Ethanol — The alcohol part of the ester.

22.4 REACTIONS OF ALCOHOLS

49. Complete combustion of an alcohol gives:

(a) CO and H_2O
(b) CO_2 and H_2
(c) CO_2 and H_2O
(d) C and H_2O

Answer: (c) CO_2 and H_2O — The reaction is highly exothermic.

50. The enthalpy of combustion of ethanol is:

(a) –890 kJ mol^{-1} (b) –286 kJ mol^{-1}
(c) +1367 kJ mol^{-1} (d) –1367 kJ mol^{-1}

Answer: (d) –1367 kJ mol^{-1} — Value given in the textbook.

51. Moles of oxygen needed to burn one mole of ethanol completely:

(a) 2 (b) 3
(c) 4 (d) 6

Answer: (b) 3 — $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$.

52. Ethanol reacts with HCl in the presence of:

(a) $ZnCl_2$ (b) Al_2O_3
(c) H_3PO_4 (d) NaOH

Answer: (a) $ZnCl_2$ — Chloroethane and water are formed.

53. In the substitution reactions of alcohols, the group replaced is:

(a) –H (b) –OH
(c) $–CH_3$ (d) –OR

Answer: (b) –OH — E.g. by a halogen.

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54. Propan-1-ol with thionyl chloride gives 1-chloropropane and:

(a) H_2O (b) H_3PO_3
(c) $POCl_3$ (d) SO_2 and HCl

Answer: (d) SO_2 and HCl — Both by-products are gases.

55. The by-product of the reaction of an alcohol with PCl_3 is:

(a) H_3PO_3 (b) $POCl_3$
(c) SO_2 (d) H_2

Answer: (a) H_3PO_3 — Phosphorous acid.

56. Ethanol reacts with sodium to give:

(a) Ethanal
(b) Ethene
(c) Sodium ethoxide and hydrogen
(d) Sodium ethanoate

Answer: (c) Sodium ethoxide and hydrogen — The O–H bond breaks.

57. The gas evolved when sodium reacts with an alcohol is:

(a) Oxygen (b) Steam
(c) Carbon dioxide (d) Hydrogen

Answer: (d) Hydrogen — Used as a test for the –OH group.

58. The reaction of sodium with ethanol, compared with water, is:

(a) More vigorous (b) Less vigorous
(c) Equally vigorous (d) Explosive

Answer: (b) Less vigorous — Ethanol is a weaker acid than water.

59. Sodium alkoxide left after evaporating excess ethanol is a:

(a) White crystalline solid (b) Yellow liquid
(c) Green solid (d) Colourless gas

Answer: (a) White crystalline solid — Stated in the textbook.

60. As the hydrocarbon chain of the alcohol becomes longer, the reaction with sodium becomes:

(a) More vigorous (b) Explosive
(c) Less vigorous (d) Unchanged

Answer: (c) Less vigorous — Stated in the textbook.

61. With sodium and phenolphthalein, ethanol turns pink:

(a) Slowly (b) Immediately
(c) Never (d) Then green

Answer: (a) Slowly — The base is formed slowly.

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62. A primary alcohol is first oxidised to:

(a) A ketone (b) An aldehyde
(c) An alkene (d) An ester

Answer: (b) An aldehyde — Which can be oxidised further to a carboxylic acid.

63. A secondary alcohol is oxidised to:

(a) An aldehyde (b) A carboxylic acid
(c) A ketone (d) An alkene

Answer: (c) A ketone — It cannot be oxidised further.

64. Which alcohol is not oxidised by acidified dichromate?

(a) Propan-1-ol (b) Ethanol
(c) Propan-2-ol (d) 2-Methylpropan-2-ol

Answer: (d) 2-Methylpropan-2-ol — It is a tertiary alcohol.

65. Acidified potassium dichromate(VI) is:

(a) Purple (b) Green
(c) Orange (d) Colourless

Answer: (c) Orange — Colour of $Cr_2O_7^{2-}$ ions.

66. When an alcohol is oxidised, dichromate ions are reduced to:

(a) Cr^{2+}
(b) Cr^{3+}
(c) Cr^{6+}
(d) CrO_4^{2-}

Answer: (b) Cr^{3+} — These ions are green.

67. The colour change with propan-1-ol and acidified dichromate is:

(a) Orange to green (b) Green to orange
(c) Purple to colourless (d) No change

Answer: (a) Orange to green — A primary alcohol is oxidised.

68. Oxidation of propan-2-ol gives:

(a) Propanal (b) Propene
(c) Propanoic acid (d) Propanone

Answer: (d) Propanone — A ketone.

69. Oxidation of ethanol first gives:

(a) Ethene (b) Ethanal
(c) Ethane (d) Propanone

Answer: (b) Ethanal — An aldehyde.

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70. Dichromate needs an acidic medium because its reduction requires:

(a) H^+ ions (b) OH^- ions
(c) Electrons only (d) Water only

Answer: (a) H^+ ions — "Keep in Mind" box.

71. Dehydration of ethanol gives:

(a) Ethane (b) Ethyne
(c) Ethanal (d) Ethene

Answer: (d) Ethene — A water molecule is removed.

72. Dehydration of alcohols is a type of:

(a) Addition (b) Substitution
(c) Elimination (d) Combustion

Answer: (c) Elimination — Water is eliminated.

73. Which is NOT a dehydrating agent for alcohols?

(a) Al_2O_3
(b) $NaBH_4$
(c) Conc. H_2SO_4
(d) Conc. H_3PO_4

Answer: (b) $NaBH_4$ — It is a reducing agent.

74. A breathalyser works on the:

(a) Combustion of ethanol (b) Dehydration of ethanol
(c) Oxidation of ethanol (d) Reduction of ethanol

Answer: (c) Oxidation of ethanol — The current is proportional to the alcohol consumed.

75. The colour of acidified $KMnO_4$ is:

(a) Orange (b) Blue
(c) Green (d) Purple

Answer: (d) Purple — It is a purple oxidising agent.

76. Butan-2-ol on oxidation gives:

(a) Butanone (b) Butanal
(c) Butanoic acid (d) But-2-ene

Answer: (a) Butanone — It is a secondary alcohol.

22.5 PHENOLS

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77. The formula of the simplest phenol is:

(a) $C_6H_5CH_2OH$
(b) $C_6H_{11}OH$
(c) C_6H_5OH
(d) $C_6H_5NH_2$

Answer: (c) C_6H_5OH — –OH is bonded directly to the ring.

78. Phenol is also known as:

(a) Picric acid
(b) Carbolic acid
(c) Benzoic acid
(d) Grain alcohol

Answer: (b) Carbolic acid — It was used in carbolic soap.

79. Phenol is a:

(a) White crystalline solid
(b) Colourless gas
(c) Green liquid
(d) Yellow solid

Answer: (a) White crystalline solid — Stated in the textbook.

80. 1-Naphthol is a derivative of:

(a) Benzene
(b) Cyclohexane
(c) Anthracene
(d) Naphthalene

Answer: (d) Naphthalene — –OH at position 1 of naphthalene.

81. Phenol can be prepared from:

(a) Phenylamine
(b) Ethanol
(c) Ethene
(d) Propanone

Answer: (a) Phenylamine — By way of a diazonium salt.

82. Nitrous acid is prepared in situ from:

(a) $NaNO_3$ and HCl
(b) HNO_3 and H_2SO_4
(c) $NaNO_2$ and HCl
(d) $NaNO_2$ and NaOH

Answer: (c) $NaNO_2$ and HCl — It is too unstable to store.

83. In the preparation of the diazonium salt the temperature is kept:

(a) Above 60 °C
(b) Below 10 °C
(c) At 100 °C
(d) At 25 °C

Answer: (b) Below 10 °C — Using ice.

84. Phenylamine reacts with nitrous acid to give:

(a) Nitrobenzene
(b) Benzene
(c) Phenyl ethanoate
(d) A diazonium salt

Answer: (d) A diazonium salt — It is unstable.

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85. The diazonium salt gives phenol on:

(a) Cooling (b) Reduction
(c) Warming (d) Drying

Answer: (c) Warming — It decomposes (is hydrolysed).

86. The formula of nitrous acid is:

(a) HNO_3 (b) HNO_2
(c) N_2O (d) NH_3

Answer: (b) HNO_2 — It is unstable.

22.6 REACTIONS OF PHENOL

87. Phenol is only slightly soluble in water because of its:

(a) –OH group (b) Large non-polar benzene ring
(c) Acidic nature (d) Low mass

Answer: (b) Large non-polar benzene ring — Stated in the textbook.

88. Phenol reacts with NaOH to give:

(a) Sodium phenoxide and water
(b) Sodium benzoate
(c) Benzene
(d) Phenyl ethanoate

Answer: (a) Sodium phenoxide and water — An acid–base reaction.

89. Molten phenol reacts with sodium to give sodium phenoxide and:

(a) Water (b) Oxygen
(c) Hydrogen (d) Benzene

Answer: (c) Hydrogen — It reacts vigorously.

90. Phenol heated with ethanoyl chloride in the presence of a base gives:

(a) Ethyl benzoate (b) Chlorobenzene
(c) Benzoic acid (d) Phenyl ethanoate

Answer: (d) Phenyl ethanoate — An ester.

91. In the reaction of phenol with an acyl chloride, the base:

(a) Deprotonates phenol (b) Oxidises phenol
(c) Reduces the acyl chloride (d) Acts as solvent

Answer: (a) Deprotonates phenol — It forms the phenoxide ion.

92. Compared with phenol, the phenoxide ion is a:

(a) Weaker nucleophile (b) Stronger acid
(c) Better nucleophile (d) Better electrophile

Answer: (c) Better nucleophile — It attacks the carbonyl carbon.

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93. The reaction of phenol with diazonium ions is discussed in the chapter on:

(a) Polymers (b) Organic nitrogen compounds
(c) Halogenoalkanes (d) Carbonyl compounds

Answer: (b) Organic nitrogen compounds — Chapter 24.

94. The –OH group of phenol makes the ring:

(a) Deactivated (b) Non-aromatic
(c) Saturated (d) Activated

Answer: (d) Activated — It donates electron density into the ring.

95. The –OH group directs incoming electrophiles to positions:

(a) 3 and 5 (b) 2, 4 and 6
(c) 3 only (d) 1 only

Answer: (b) 2, 4 and 6 — Stated in the textbook.

96. Phenol with dilute nitric acid at room temperature gives:

(a) 3-Nitrophenol (b) Nitrobenzene
(c) 2- and 4-nitrophenol (d) 2,4,6-Trinitrophenol

Answer: (c) 2- and 4-nitrophenol — A mixture of the two isomers.

97. Phenol with concentrated nitric acid gives:

(a) 2,4,6-Trinitrophenol (b) 2-Nitrophenol only
(c) 4-Nitrophenol only (d) Nitrobenzene

Answer: (a) 2,4,6-Trinitrophenol — All three positions are substituted.

98. In nitration, a hydrogen atom of the ring is replaced by:

(a) –OH (b) $-SO_3H$
(c) $-NH_2$ (d) $-NO_2$

Answer: (d) $-NO_2$ — The nitro group.

99. Phenol with bromine water gives:

(a) Bromobenzene (b) 2,4,6-Tribromophenol
(c) 3-Bromophenol (d) No reaction

Answer: (b) 2,4,6-Tribromophenol — A white precipitate at room temperature.

100. The bromination of phenol with bromine water requires:

(a) No catalyst (b) $AlBr_3$ catalyst
(c) UV light (d) Heating at 60 °C

Answer: (a) No catalyst — The ring is activated.

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101. The bromination of benzene needs:

(a) Bromine water only
(b) NaOH
(c) Dilute HNO_3
(d) Pure bromine and anhydrous $AlBr_3$

Answer: (d) Pure bromine and anhydrous $AlBr_3$ — Benzene is less reactive than phenol.

102. Benzene is nitrated with:

(a) Dilute HNO_3
(b) $NaNO_2$ and HCl
(c) Conc. HNO_3 and conc. H_2SO_4
(d) Bromine water

Answer: (c) Conc. HNO_3 and conc. H_2SO_4 — Refluxed between 25 °C and 60 °C.

103. The reaction of the phenol ring with bromine is:

(a) Nucleophilic substitution (b) Electrophilic substitution
(c) Electrophilic addition (d) Elimination

Answer: (b) Electrophilic substitution — An electrophile attacks the electron-rich ring.

104. 1-Naphthol with bromine in ethanoic acid gives:

(a) 3-Bromo-1-naphthol
(b) Bromobenzene
(c) 9-Bromo-1-naphthol
(d) 2-Bromo- and 4-bromo-1-naphthol

Answer: (d) 2-Bromo- and 4-bromo-1-naphthol — –OH directs to positions 2 and 4.

105. Substitution at positions 9 and 10 of 1-naphthol is not possible because these carbons:

(a) Are too reactive
(b) Carry the –OH group
(c) Have no hydrogen atom
(d) Are saturated

Answer: (c) Have no hydrogen atom — They are shared by both rings.

106. The test for phenol is the formation of a white precipitate with:

(a) Bromine water (b) Sodium
(c) Acidified dichromate (d) Sodium hydroxide

Answer: (a) Bromine water — The orange colour is also decolourised.

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